

Lab Exercise

1. Calculate the total resistance in ohms.

i. Total resistance in series (a)

$$\text{Resistance} = 20 + 30 + 50$$

$$\text{Total resistance} = \mathbf{100\ \Omega}$$

ii. Total resistance for the parallel circuit (b)

$$1/\text{Total resistance} = (1/20) + (1/100) + (1/50)$$

$$= 0.05 + 0.01 + 0.02 = 0.08$$

$$\text{Total resistance} = 1/0.08 = \mathbf{12.5\ \Omega}$$

2. Calculate the total current from the power supply.

i. Total current in the series circuit (a)

$$V = IR$$

$$I = V/R$$

$$I = 125/100$$

$$\mathbf{I = 1.25\ A}$$

ii. Total current in the parallel circuit (b)

$$I = V/R$$

$$I = 125/12.5$$

$$\mathbf{I = 10\ A}$$

3. Calculate the current through each resistor

i. Current through each resistor in the series circuit (a)

In a series circuit, current is the same everywhere:

$$I \text{ at } 20\ \Omega = \mathbf{1.25\ A}$$

$$I \text{ at } 30\ \Omega = \mathbf{1.25\ A}$$

$$I \text{ at } 50\ \Omega = \mathbf{1.25\ A}$$

ii. Current through each resistor in the parallel circuit (b)

$$I = V/R$$

$$\text{For } 20\ \Omega = 125/20 = \mathbf{6.25\ A}$$

For $100\ \Omega = 125/100 = \mathbf{1.25\ A}$

For $50\ \Omega = 125/50 = \mathbf{2.5\ A}$

4. Calculate the Voltage drop through each resistor.

i. **Voltage drop in the series circuit (a)**

$$V = IR$$

$$V \text{ at } 20\ \Omega = 1.25 * 20 = \mathbf{25\ V}$$

$$V \text{ at } 30\ \Omega = 1.25 * 30 = \mathbf{37.5\ V}$$

$$V \text{ at } 50\ \Omega = 1.25 * 50 = \mathbf{62.5\ V}$$

ii. **Voltage drop in the parallel circuit (b)**

Voltage is the same in parallel:

$$V \text{ at } 20\ \Omega = \mathbf{125\ V}$$

$$V \text{ at } 100\ \Omega = \mathbf{125\ V}$$

$$V \text{ at } 50\ \Omega = \mathbf{125\ V}$$

5. Calculate the Power dissipated through each resistor (use Watt's Law: Power = Voltage * Current)

i. **Power dissipated in the series circuit (a)**

$$P = VI$$

$$P \text{ at } 20\ \Omega = 25 * 1.25 = \mathbf{31.25\ W}$$

$$P \text{ at } 30\ \Omega = 37.5 * 1.25 = \mathbf{46.88\ W}$$

$$P \text{ at } 50\ \Omega = 62.5 * 1.25 = \mathbf{78.13\ W}$$

ii. **Power dissipated in the parallel circuit (b)**

$$P = VI$$

$$P \text{ at } 20\ \Omega = 125 * 6.25 = \mathbf{781.25\ W}$$

$$P \text{ at } 100\ \Omega = 125 * 1.25 = \mathbf{156.25\ W}$$

$$P \text{ at } 50\ \Omega = 125 * 2.5 = \mathbf{312.5\ W}$$